



Detection of Insect-Damaged Maize Seed Using Hyperspectral Imaging and Hybrid 1D-CNN-BiLSTM Model

Zheli Wang^{a,b}, Shuxiang Fan^b, Ting An^b, Chi Zhang^b, Liping Chen^{a,b,*}, Wenqian Huang^b

^a College of Information and Electrical Engineering, China Agricultural University, Beijing 100083, China

^b Intelligent Equipment Research Center, Beijing Academy of Agriculture and Forestry Sciences, Beijing 100097, China

ARTICLE INFO

Keywords:

Maize seed
Insect infestation
Hyperspectral imaging
Deep learning
BiLSTM

ABSTRACT

Maize (*Zea mays* L.) holds a pivotal position in various domains, making maize seeds' germination rate and quality crucial for agricultural production. Insect-infested seeds can impair their germination rate, resulting in substantial economic losses. Nevertheless, research on rapid detection methods for insect infestation in maize seeds remains insufficient. In this study, we employed hyperspectral data within the range of 930–1866 nm alongside intelligent algorithms to address the classification of healthy and insect-infested maize seeds. The 1D-CNN-BiLSTM and SVM models were applied to diverse combinations of spectra and texture features. The results indicated that the 1D-CNN-BiLSTM model exhibited clear advantages over SVM in handling multi-source data. Additionally, ANOVA identified the 1160 nm / 1310 nm band ratio as optimal. The 1D-CNN-BiLSTM model, utilizing texture features from this band ratio image, exhibited exceptional classification performance with just two band images. The GLCM + 1D-CNN-BiLSTM model garnered the best results, with the F1-score and accuracy of 0.96, respectively. This approach streamlined the band number, reducing model complexity and rendering it promising for practical applications. Consequently, the results of this study provide new insights and methods for classifying insect infestation maize seeds.

1. Introduction

Maize (*Zea mays* L.) ranks among the most vital cereal crops globally, serving as a staple food for millions of people and a vital feed source for livestock. With its versatile applications in various industries, including food, biofuels, and industrial products, the demand for high-quality maize seeds has steadily increased. The germination capacity and quality of maize seeds are pivotal in determining crop yield and overall agricultural productivity [1].

However, one of the significant challenges faced by the agricultural industry is the infestation of maize seeds by insects. Insect damage can harm seed viability, reducing germination rates, stunted seedling growth, and ultimately decreasing crop yields [2]. Moreover, insect infestation seeds can contribute to the spread of pests to healthy seeds, causing widespread agricultural losses. Early detection and removal of insect infestation seeds are essential for maintaining seed quality and crop productivity. Traditionally, manual inspection has been employed to identify damaged seeds, which is time-consuming and labor-intensive. Additionally, manual assessment may not be sufficiently accurate, potentially resulting in the misclassification of healthy and

insect-infested seeds.

In recent years, non-destructive seed quality assessment methods have gained traction as they offer the advantage of preserving seed integrity while providing valuable information about their internal attributes. Among these methods, computer vision and near-infrared spectroscopy (NIRS) technology are two prominent directions for non-destructive quality inspection. Computer vision techniques involve the analysis of seed images to identify external defects, such as color variations or surface irregularities. Lin, Shu, Zhong, Lu, Ma and Meng [3] proposed a computer vision-based method for online classification of soybean seeds, and results indicated that the F-score of the normal, damaged, abnormal, and non-classification soybeans reached a high level. Li, Zhao, Wu, Zhang, Lv and Li [4] used the machine-vision-based method for stress-crack detection in maize kernels, and the observed precision and recall of the overall system were 92.7 % and 94.4 %, respectively. Tu, Wu, Cheng, Zhang, Xu, Dong, Wang and Sun [5] developed a novel software named Alseed that automatically captured and analyzed great traits, such as the shape, color, or texture features of individual seeds, which could be used for seed quality detection and prediction such as seed purity, vigor, and variety. Although these

* Corresponding author at: College of Information and Electrical Engineering, China Agricultural University, Beijing 100083, China.

E-mail address: chenlp@nrcita.org.cn (L. Chen).

<https://doi.org/10.1016/j.infrared.2024.105208>

Received 29 December 2023; Received in revised form 29 January 2024; Accepted 1 February 2024

Available online 7 February 2024

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methods are relatively straightforward, they may not capture subtle internal changes caused by insect infestations. Thus, computer vision techniques may be limited in accurately distinguishing healthy and insect infestation seeds. NIRS technology is a non-destructive detection method that can acquire spectral information of samples without causing damage [6]. This helps to maintain the integrity and usability of seeds. This technology enabled the prediction of various parameters from seeds, such as cracked seed [7], protein content [8–10], oil content [11], and haploids [12], without compromising the seed's quality. This comprehensive approach enhances the accuracy of seed quality assessment. Additionally, NIRS eliminates the need for sample preparation, reducing operational costs and time. In our preliminary research, we developed a near-infrared rapid detection system capable of achieving fast detection of insect infestation and protein content [2]. However, the development of NIRS equipment for seed quality assessment is a complex process, often necessitating customized devices for different seed varieties. Moreover, the heterogeneity of samples can result in spectral data variations, affecting models' stability and generalization ability.

Hyperspectral imaging technology has emerged as a powerful tool for seed quality inspection. It combines the advantages of image and spectra analysis, enabling the acquisition of detailed spectral and spatial information from each seed [13]. Hyperspectral imaging offers a non-destructive and non-invasive approach to examine internal seed properties, including biochemical composition and physiological characteristics [14]. Aulia, Kim, Zuhrotul Amanah, Muhammad Akbar Andi, Kim, Kim, Lee, Kim, Baek and Cho [15] applied near-infrared hyperspectral imaging for protein content prediction in single soybean seeds. Guo, Zhang, Ma, Yin, Guo, Sun and Jin [16] used the hyperspectral imaging system to achieve moisture content detection in soybean seeds. Moreover, many researchers also employed the hyperspectral imaging system in vigor detection [17–19], oil prediction [20], toxin detection [21] and hardness determination [22]. From the current research, the application of hyperspectral of hyperspectral technology in maize quality inspection offers several benefits. Firstly, it enables early and accurate detection of damaged seeds, minimizing economic losses for farmers and improving crop yields. Secondly, the non-destructive nature of hyperspectral imaging ensures that seed integrity remains intact, enabling seeds to be used for planting after inspection. However, some challenges are still associated with detection methods based on hyperspectral imaging technology. Firstly, current research predominantly relies on traditional machine learning methods, with the application of deep learning in the hyperspectral domain being relatively limited. Secondly, the substantial volume of hyperspectral data poses difficulties constructing models suitable for rapid detection. Thirdly, few researchers utilized multi-source data from hyperspectral information, especially concerning small-sample scenarios like maize seeds. Lastly, few researches focused on insect infestation detection in seeds using hyperspectral imaging. Therefore, it is essential to develop a rapid detection model for insect infestation in maize seeds based on multi-source hyperspectral data. This model should ensure accuracy while using as few hyperspectral data as possible to achieve rapid model building and deployment.

This research aimed to explore the potential of hyperspectral imaging technology to detect insect-infestation seeds. Specific objectives were to (1) collect spectral data from full band hyperspectral images and texture data from dimensionally reduced hyperspectral images, (2) perform linear and nonlinear dimensionality reduction on the texture data, (3) construct deep learning models using various combinations of spectral and texture features, (4) optimize the models based on ANOVA method.

2. Materials and methods

2.1. Sample preparation and data acquisition

The variety of maize seeds used in this research was JINGKE968. The total number of samples was 240, with an equal distribution of 50 %

healthy seeds and 50 % damaged seeds. 168 randomly selected samples were used as the calibration set to establish the discriminative model. In contrast, the remaining 72 seeds were used to test the performance of the model.

The hyperspectral imaging system with a near-infrared range (930–2548 nm, S) was used to acquire hyperspectral images of the embryo side of maize seeds. The reason for collecting hyperspectral data on maize seed embryo side is that the embryo is more susceptible to damage due to their relative vulnerability to nibbling compared to the endosperm. Furthermore, it has been observed that embryo is a vulnerable area. Typically, even when the embryo is completely nibbled, the endosperm can remain intact. The wavelength resolution of the system is 6.32 nm, thus resulting in a total of 256 bands. Consequently, the dimensions of the captured raw hyperspectral image are 320 pixels in the x-direction, a variable number of pixels in the y-direction depending on the sample length, and 256 bands in the z-direction. In order to enhance the collection efficiency, every 60 maize seeds were placed on a dark-background sampling plate for the data collection. Therefore, a total of four collections were conducted to complete the data collection of all samples. When collecting hyperspectral data, the exposure time is set to 3 ms, and the platform moves at a speed of 25 mm/s. Furthermore, the seeds with different labels were alternately placed within each group to reduce the equipment and system errors caused by different collection batches. The specific procedure is to place 5 healthy seeds in the first row and 5 Insect-Damaged seeds in the second row, alternating in this manner. The data acquisition process is shown in Fig. 2. The non-uniformity intensity of light in different bands and the presence of dark currents can increase noise in certain bands. Thus, the original hyperspectral images should be corrected by the white and dark references. The following equation calculates the corrected image:

$$R_c = \frac{R_{raw} - R_{dark}}{R_{white} - R_{dark}} \quad (1)$$

where R_c and R_{raw} represent the corrected and raw images, R_{dark} and R_{white} represent the dark and white references, respectively.

2.2. Texture extraction methods

The texture of sample images usually reflects the biophysical and biochemical characteristics directly or indirectly correlated with the quality of samples. The advantage of hyperspectral image technology is the integration of image and spectrum technology, which can be more convenient to obtain various data. Therefore, the hyperspectral images were also used to extract the texture feature in this study. Due to the large volume of hyperspectral image data, this study employed principal component analysis for dimensionality reduction on the hyperspectral image data. Texture data was then extracted from the first three principal component images, respectively.

Gray-level co-occurrence matrix (GLCM), a classic texture extraction method, has been widely applied in the field of agricultural product quality detection for capturing texture patterns and relationships within images [23–25]. GLCM calculates the frequency of pixel intensity value pairs occurring at a specified distance and in a specific direction within an image. The resulting matrix, often square and symmetric, captures the distribution of pixel-intensity relationships. Once the GLCM matrix is constructed, various texture features can be derived. In this study, six texture features (contrast, dissimilarity, homogeneity, energy, correlation, and angular second moment (ASM)) were derived from the GLCM for insect infestation detection. The texture features were extracted from four independent directions (0, 45, 90, and 135) at a distance of 1 and gray levels of 256. Then, the average value of each texture in the four directions was used as the final value of each image. Thus, each image could extract six texture features. Each texture feature formula is as follows:

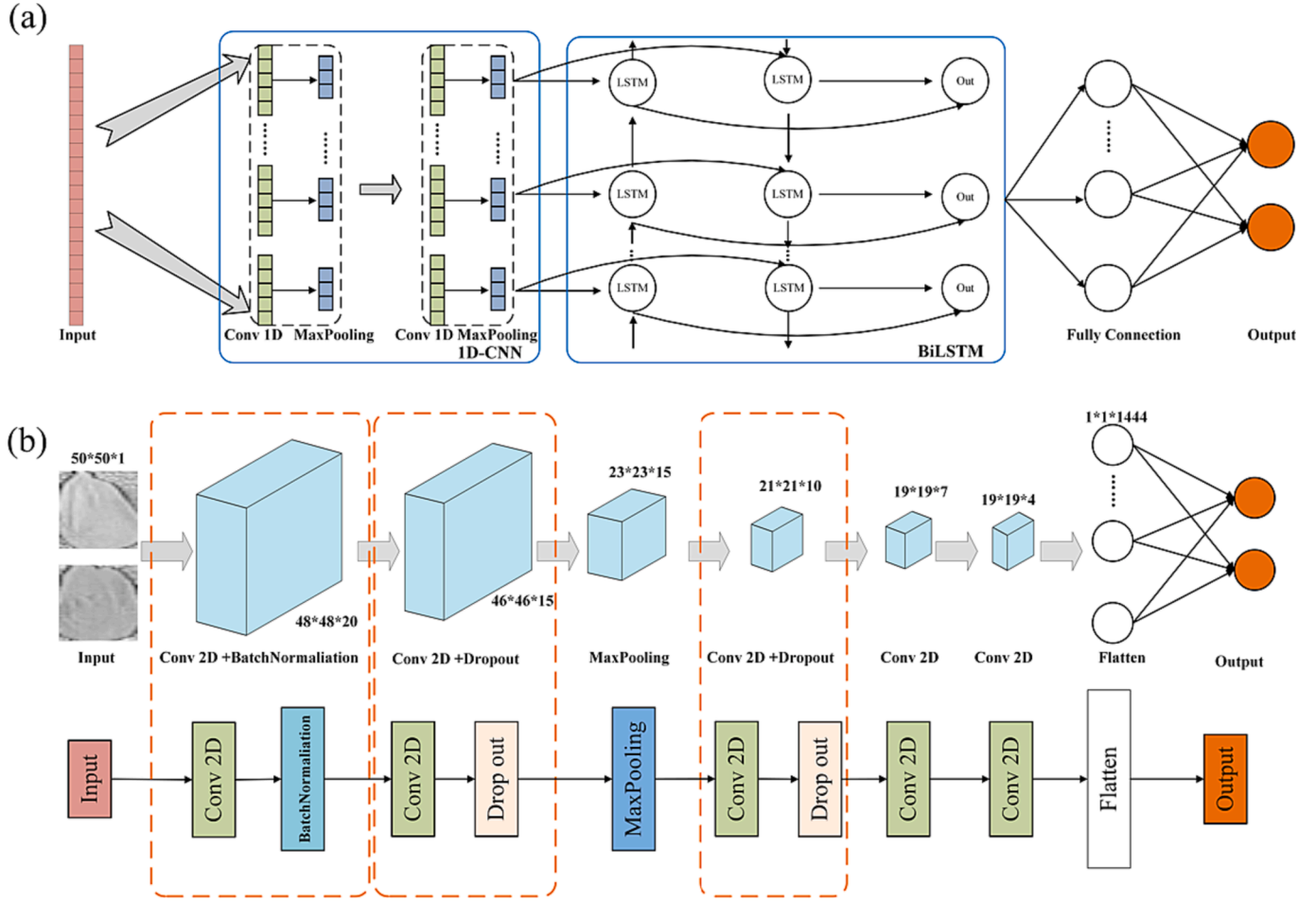


Fig. 1. The convolutional neural network topology. (a) 1D-CNN-BiLSTM; (b) 2D-CNN.

$$Contrast = \sum_{i,j=0}^{N-1} (i-j)^2 P(i,j) \quad (2)$$

$$Dissimilarity = \sum_{i,j=0}^{N-1} |i-j| P(i,j) \quad (3)$$

$$Homogeneity = \sum_{i,j=0}^{N-1} \frac{P(i,j)}{1 + (i-j)^2} \quad (4)$$

$$ASM = \sum_{i,j=0}^{N-1} (P(i,j))^2 \quad (5)$$

$$Correlation = \sum_{i,j=0}^{N-1} P(i,j) \left[\frac{(i-\mu_i)(j-\mu_j)}{\sqrt{(\sigma_i^2)(\sigma_j^2)}} \right] \quad (6)$$

$$Energy = \sqrt{ASM} \quad (7)$$

Where i and j represent the values of the gray levels; N represents the gray level; μ_i , μ_j , and σ_i , σ_j are the means and standard deviations, respectively, of rows and columns in the GLCM matrix. $P(i, j)$ is the probability of corresponding (i, j) element in the GLCM matrix.

Local binary pattern (LBP) is a commonly employed technique in computer vision for analyzing and describing texture patterns. LBP operates by comparing the intensity value of a central pixel with its surrounding neighbor pixels. It defines a binary code for each pixel based on whether the neighboring pixels have higher or lower intensity

values than the central pixel [26]. This binary code is then converted into a decimal value, representing the local pattern at that pixel. The process is applied to each pixel in the image to obtain the LBP descriptor for an image. The resulting LBP values create a histogram representing the distribution of different local patterns with the image. This histogram can be used as a feature vector to characterize the texture of the image. In this study, the number of neighboring pixels was set to 8, the radius of the neighborhood was 1, and the encoding method for LBP was 'uniform', 10 texture features could be extracted from each image by using LBP.

Histogram of oriented gradient (HOG), a widely used feature extraction technique in image processing, aims to capture and describe the local gradient patterns in an image for object detection and recognition tasks [27–29]. HOG operates by dividing an image into small regions called cells. For each cell, the gradient magnitude and orientation of pixels within the cell are computed. The gradient magnitude represents the local intensity variations, while the gradient orientation indicates the direction of these variations. The cells are further grouped into larger blocks to capture the spatial arrangement of gradient orientations. The histogram of gradient orientations is computed for each block, representing the distribution of gradient orientations within the block. This histogram is often normalized to enhance the robustness against illumination variations. Finally, the HOG feature vector is constructed by concatenating the histograms of all the blocks in the image. In this research, the image size was resized to 45*45, the pixels per cell were set to 8*8, and each block consisted of 2*2 cells. The block normalization was performed using L2-Hys, a technique that enhances the robustness of the gradient histograms and the resulting features against variations in illumination.

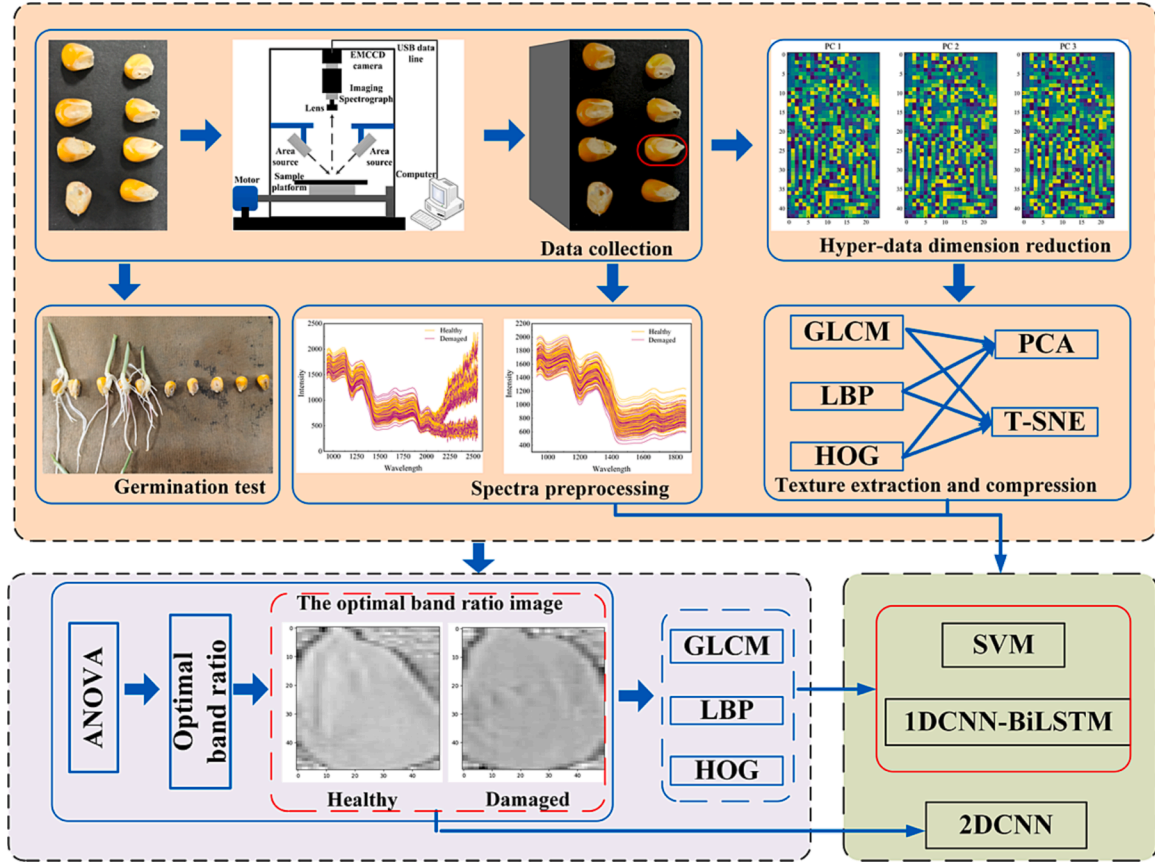


Fig. 2. The flowchart of this research.

In summary, GLCM is suitable for capturing global texture structures, LBP is suitable for emphasizing local texture features, and HOG is suitable for detecting texture features with clear orientation. Thus, these three methods can provide a more comprehensive description of texture information in images.

2.3. Dimension reduction method

Principal component analysis (PCA) is a linear technique used to reduce the dimensionality of data by transforming high-dimensional data into a lower-dimensionality of data by transforming high-dimensional data into a lower-dimensional space while retaining essential information [30]. By using this method, the processed spectral data are decomposed into several principal components (PCs), which are linear combinations of the original spectral data [31]. The first principal component captures the most significant variability in the data, followed by subsequent components in decreasing order of importance. PCA is widely used for data visualization, noise reduction, feature extraction, and compression. PCA also has various applications in different domains, including image and signal processing and pattern recognition [32,33]. In this study, PCA was applied twice for data dimensionality reduction. The first time was to compress the high-dimensional hyper-spectral image data into three dimensions, facilitating the extraction of texture features and improving processing speed. The second use of PCA was to purify the multidimensional texture data, achieving for linear dimensionality reduction of the texture features.

T-distributed stochastic neighbor embedding (T-SNE), a nonlinear dimensionality reduction technique, focuses on preserving the local structure and relationships between data points [34]. It is particularly effective in visualizing high-dimensional data in two or three dimensions, revealing clusters, patterns, and similarities that may be hidden in the original space. Unlike PCA, T-SNE emphasizes the

preservation of pairwise distances or similarities between data points rather than global variance. The core concept of T-SNE involves constructing a probability distribution in the high-dimensional space to represent pairwise similarities between data points and creating a similar probability distribution in the lower-dimensional space simultaneously. The objective is to minimize the Kullback-Leibler divergence between these two distributions, ensuring that similar points are modeled by similar distributions in the lower-dimensional embedding. The algorithm iteratively adjusts the positions of the points in the lower-dimensional space until a stable representation is achieved. T-SNE finds applications in visualizing complex datasets, clustering analysis, anomaly detection, and exploratory data analysis. It has proven effective in various domains, including natural language processing and image recognition [35–37].

Analysis of variance (ANOVA), being one of the most robust and widely employed statistical methods for group comparison, was used in this study to examine the variations existing between different groups [38]. Specifically, it was utilized to identify the optimal band ratio for distinguishing between labels. The F-values derived from the ANOVA analysis were employed to select the features that exhibited statistically significant differences across the various labels. The band ratio associated with the highest F-value indicated the presence of the most substantial differences between the different groups when considering this specific band ratio. In the study, the ANOVA method was employed to analyze the spectral data in order to obtain the band ratios that best reflect differences between different labels and subsequently determining the optimal threshold. Additionally, band ratio images can be extracted based on these ratios for further analysis in the subsequent steps.

2.4. Model establishment methods

Support vector machine (SVM) is a powerful machine learning algorithm that is widely employed in binary classification tasks [39]. The fundamental principle of SVM is to find an optimal hyperplane that can separate different classes of data points with the maximum margin. The hyperplane is the decision boundary that maximizes the distance between the nearest data points from different classes, known as support vectors. By maximizing this margin, SVM aims to improve generalization and classification performance on unseen data. The SVM algorithm can handle both linearly separable and non-linearly separable datasets. For linearly separable data, SVM employs a linear kernel function to transform the input data into a higher-dimensional feature space, where the classes can be linearly separated. In the case of non-linearly separable data, SVM utilizes a kernel trick to implicitly map the data into a higher-dimensional space, where a linear hyperplane can effectively separate the classes. The SVM algorithm was employed as a benchmark method to assess the performance of other modeling approaches in this study.

In recent years, deep learning has been applied in spectral non-destructive detection [40,41]. In this research, the 1D-CNN-BiLSTM model was proposed for one-dimensional data. The 1D-CNN-BiLSTM is a neural network architecture that combines 1D convolutional neural networks (CNNs) and bidirectional long short-term memory (BiLSTM) layers. 1D-CNN is a variant of CNN specifically designed to process one-dimensional input sequences. It applies convolutional operations to capture local patterns and features from the input sequence. The 1D convolutional layers have filters that slide across the input sequence, extracting relevant features at different positions. On the other hand, BiLSTM is a type of recurrent neural network (RNN) designed to process sequences bidirectionally. It comprises two long short-term memory (LSTM) layers, one processing the sequence in the forward direction, and the other in the backward direction. The LSTM architecture consists of a cell unit and three gates, which are the input gate, forget gate, and output gate [42]. It could resolve the gradient disappearing and explosion problem during long sequence training [43]. By combining 1D-CNN and BiLSTM layers, the network can leverage the strengths of both architectures. The 1D-CNN layers can effectively extract local patterns and features from the input sequence, while the BiLSTM layers can capture long-range dependencies and context information. This combination enables the model to learn more comprehensive and discriminative information from sequential data. The architectures of 1D-CNN-BiLSTM used in this research are shown in Fig. 1 (a).

In order to investigate a simpler, faster, and more practical model for application, the 2D-CNN model was also developed based on band images. However, hyperspectral data consists of many band images, which poses challenges for building deep convolutional models. Therefore, it is crucial to select band images relevant to insect infestation for building a deep learning model. In this research, the ANOVA was used to determine the optimal band ratio. Then, the optimal band ratio image can be selected for constructing the classification model. The architectures of 2D-CNN used in this research are shown in Fig. 1 (b).

2.5. Model evaluation metrics

In this study, the precision, accuracy, recall, and F1-score were used to evaluate the model performance. Precision measures the ratio of correctly predicted positive instances to all instances predicted as positive. It focuses on the accuracy of positive prediction. A high precision indicates a low false positive rate, meaning that the model is precise in identifying positive instances. Accuracy measures the overall correctness of the model's predictions by comparing the correctly predicted samples to the total number of samples. It calculates the ratio of correctly predicted samples (both positive and negative) to the total number of samples. Recall, also known as sensitivity or true positive

rate, measures the proportion of correctly predicted positive samples out of all actual positive samples in the dataset. A high recall indicates a low false negative rate, meaning the model can effectively capture positive samples. The F1-score is a popular evaluation metric that combines precision and recall into a single value. The F1-score is calculated as the harmonic mean of precision and recall and provides a balanced measure of a model's performance. The harmonic mean gives more weight to lower values. The formulas for precision, accuracy, recall, and F1-score are as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives. In this paper, the healthy samples were represented as positive, and the damaged samples were regarded as negative.

2.6. Software tool

Python was used for machine learning model development, data dimensionality reduction and texture extraction, while graph construction was carried out using Origin 2018 software (Origin Lab Corporation, Northampton, MA, USA). All software operations were performed on a Win 10 64-bit operating system, powered by an Intel® Core™ i5-8300H CPU at 2.30 GHz and equipped with 8 GB RAM.

3. Results and discussion

3.1. Spectra analysis

The spectra curves of healthy and damaged seeds are presented in Fig. 2. It is clear that the signal-to-noise ratio of spectra decreased significantly after 1866 nm. Furthermore, the system error of the acquisition equipment surpassed the sample differences after 1866 nm, leading to obvious abnormalities in the spectra curves. Therefore, the spectral range of 930–1866 nm (total of 150 bands) was selected for study (S1). NIR spectroscopy exhibits heightened sensitivity to molecular bonds such as O–H, C–H, and N–H. It is clear that peaks and valleys manifest at approximately 980 nm, 1110 nm, 1200 nm, and 1440 nm. The valley observed near 980 nm may be attributed to second overtone of O–H bonds present in water molecules and carbohydrate structures [44,45]. Concomitantly, the absorption features discerned around 1100 nm and 1200 nm are associated with second overtone of C–H stretching vibrations within carbohydrate [46]. Moreover, the valley around 1440 nm correlates with the first overtone of O–H and N–H stretching vibrations [47,48].

However, the spectra curves of two types exhibited similar trends with comparable peaks and valleys. In addition, it can be seen clearly that the curves of healthy and damaged samples overlap, making it difficult to classify them directly based on spectral values. This suggests that there is no significant change in the chemical composition of healthy and damaged seeds, and insect infestation belongs to physical damage. Hence, it is essential to develop intelligent algorithms to classify the healthy and damaged seeds.

3.2. Germination test results

After collecting hyperspectral images for all maize seeds, the standard germination tests were conducted for all samples. The germination experiment followed the guidelines established by the International Seed Testing Association [49]. In this research, the 'between paper (BP)' method was employed for germination. The seeds were positioned on moistened germination paper, and the germination test was conducted inside a standard germination box under constant illumination at 25°C for seven days. Seeds with a germ length of 1 cm or longer were considered germinated.

It is clear from the germination results that seeds affected by insect infestation do not germinate properly, whereas healthy seeds exhibit normal germination (can be seen in Fig. 2). This difference can be attributed to the fact that insect infestation specifically affects the embryos of maize seeds, which are essential for germination. Furthermore, upon examining the samples, it was observed that the damage is situated on the embryo side, with a hole at the top of the embryo resulting from insect feeding. Nevertheless, when observing the endosperm side, the damaged seed appears identical to healthy samples. As a result, this type of damage is complicated to detect and requires manual selection one by one, which is time-consuming and labor-intensive. Therefore, if the damaged seeds are not removed before planting, it will severely impact the crop yield. Therefore, it is necessary to detect the insect infestation seeds quickly and non-destructively.

3.3. The classification results based on SVM and 1D-CNN-BiLSTM

3.3.1. The results of texture extraction from principal component images

Extracting texture features from all band images in hyperspectral data is not always optimal. The reason is that hyperspectral images have a large amount of data, resulting in redundant texture features. Furthermore, not all images are suitable for extracting texture features. For example, some band images with low signal-to-noise ratio contain significant noise, and texture features extracted from such images would degrade the performance of the model. Therefore, PCA was employed to perform dimensionality reduction on the original hyperspectral image. Texture features were extracted from the first, second, and third principal component images. Thus, the number of texture features of GLCM, LBP, and HOG were 18 (6*3), 30 (10*3), and 1728 (576*3).

Texture data may exhibit both linear and nonlinear relationships with sample labels. Therefore, PCA and T-SNE methods were further utilized for data distillation of the texture data after extracting GLCM, LBP, and HOG textures from first three principal component images. The linear dimensionality reduction was performed using PCA, reducing the data to 10 dimensions. Nonlinear dimensionality reduction was achieved using T-SNE, reducing the data to 3 dimensions. In this section, the texture feature number of GLCM, LBP, and HOG were 13 (10 dimensions by PCA and 3 dimensions by T-SNE), respectively, after dimensionality reduction. In addition, the compressed texture features were normalized (min-max normalized) to [0, 1] range. This normalization ensures that all texture features are on a comparable scale, which is particularly important in analysis as it involves a variety of texture features that originally had different ranges and units. By normalizing these features, we were able to mitigate potential biases and enhance the consistency of our dataset, facilitating more reliable and interpretable machine learning analyses.

This study integrated the spectra data with different texture data separately. The combination of spectral and texture data is achieved by concatenating the spectral data matrix and the texture data matrix into a single matrix. Then, the machine learning model (SVM) and deep learning method (1D-CNN-BiLSTM) were established to explore the optimal data fusion approach and classification algorithms. Furthermore, prior to constructing the models, we normalized the data to standardize the numerical ranges of various features, thereby preventing the impact of scale variations among feature values on the models. This

ensures that each feature operates within the same numerical range, providing a more stable and reliable data input, thereby facilitating algorithm convergence and enhancing the efficiency and accuracy of model training.

3.3.2. The classification results based on the combination of spectra and texture

The Table 1 presents the results of various machine learning models (SVM and 1D-CNN-BiLSTM) applied to different combinations of spectra and texture features (S, S1, S1 + LBP, S1 + HOG, S1 + GLCM, S1 + LBP + GLCM, S1 + LBP + HOG, S1 + HOG + GLCM, S1 + LBP + GLCM + HOG). It can be found that the purified spectra (S1) could obtain a better performance with a higher accuracy and F1-score in prediction set. This also showed the rationality spectra pretreatment. It is clear that both the SVM and 1D-CNN-BiLSTM exhibited promising performance across different combinations of spectra and texture features. The SVM model achieved precision ranging from 0.86 to 0.94, recall from 0.89 to 0.97, F1-score from 0.90 to 0.96, and accuracy from 0.90 to 0.97. Similarly, the 1D-CNN-BiLSTM maintained competitive results, with precision values ranging from 0.89 to 0.94, recall from 0.91 to 0.97, F1-score from 0.90 to 0.96, and accuracy from 0.90 to 0.97.

Notably, the performance of SVM models was superior to that of 1D-CNN-BiLSTM models when dealing with spectra data. When spectra data was combined with a single texture, both SVM and 1D-CNN-BiLSTM models showed comparable performance. This result indicated that the SVM model had an advantage in handling relatively single-source data. However, the 1D-CNN-BiLSTM models demonstrated stronger predictive ability compared to SVM models when the spectra data was combined with multiple textures. The 1D-CNN-BiLSTM models achieved higher recall compared to the SVM model, indicating its ability to capture more positive instances accurately. Additionally, this model achieved a higher F1-score, which implied a better balance between precision and recall, demonstrating its superiority in classification accuracy on the prediction set. The results show that the 1D-CNN-BiLSTM models were more effective in handling the complexities and interactions among multiple texture features, resulting in improved classification accuracy. This advantage can be attributed to the model's ability to capture both spatial and temporal dependencies within the data. The 1D-CNN of the model was proficient in extracting local features and spatial patterns from the multi-source data. At the same time, the BiLSTM enabled the model to capture temporal dependencies and patterns effectively. The combination allowed the 1D-CNN-BiLSTM model to leverage the diverse and interactive information in different texture and spectra data, leading to superior performance in handling multi-source data.

Combining spectra and texture features (S1 + LBP + GLCM) with the 1D-CNN-BiLSTM model consistently produced the highest F1-score and accuracy of 0.96, respectively. The data demonstrated the advantages of 1D-CNN-BiLSTM in dealing with multi-source data. This superior performance can also be attributed to the complementary nature of LBP and GLCM texture features. LBP effectively captures local patterns and texture variations in images, while GLCM is adept at characterizing spatial relationships between pixel intensities. When combined, LBP and GLCM provide a comprehensive representation of both local and spatial texture information, which can enhance the discriminative power of the model. On the other hand, adding HOG texture resulted in a decrease in performance. This could be that HOG is primarily designed for object detection and recognition in images, and it may not be as effective in capturing texture patterns in this specific classification problem. Furthermore, HOG may introduce additional noise and redundancy into the feature space, decreasing classification accuracy. Hence, for this particular texture-based classification task, combining LBP and GLCM with spectra data (S1) may be the most effective, as it captures both local and spatial texture information without introducing irrelevant features. This combination provides richer and more informative data, resulting in improved classification performance.

Table 1

The classification results of SVM and 1D-CNN-BiLSTM model based on texture and spectra data.

Data	Model	Calibration set				Prediction set			
		Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy
S	SVM	0.97	0.98	0.97	0.97	0.87	0.94	0.90	0.90
	BiLSTM	0.99	0.96	0.98	0.98	0.97	0.80	0.88	0.89
S1	SVM	0.91	0.96	0.94	0.93	0.89	0.97	0.93	0.93
	BiLSTM	0.94	0.98	0.96	0.96	0.87	0.97	0.92	0.92
S1 + LBP	SVM	0.96	0.96	0.96	0.96	0.92	0.94	0.93	0.93
	BiLSTM	0.93	0.95	0.94	0.94	0.89	0.97	0.93	0.93
S1 + HOG	SVM	0.98	0.99	0.98	0.98	0.89	0.94	0.92	0.92
	BiLSTM	0.99	0.94	0.96	0.96	0.89	0.91	0.90	0.90
S1 + GLCM	SVM	0.95	0.98	0.97	0.96	0.92	0.97	0.94	0.94
	BiLSTM	0.90	1.00	0.95	0.95	0.87	0.97	0.92	0.92
S1 + LBP + GLCM	SVM	0.98	0.98	0.98	0.98	0.92	0.94	0.93	0.93
	BiLSTM	0.98	0.99	0.98	0.98	0.94	0.97	0.96	0.96
S1 + LBP + HOG	SVM	0.97	0.99	0.98	0.98	0.89	0.94	0.92	0.92
	BiLSTM	0.98	0.94	0.96	0.96	0.89	0.97	0.93	0.93
S1 + HOG + GLCM	SVM	0.98	0.98	0.98	0.98	0.89	0.94	0.92	0.92
	BiLSTM	1.00	0.93	0.96	0.96	0.97	0.91	0.94	0.94
S1 + LBP + GLCM + HOG	SVM	0.98	0.96	0.97	0.97	0.94	0.86	0.90	0.90
	BiLSTM	1.00	0.95	0.98	0.98	0.97	0.91	0.94	0.94

Abbreviation: S: The raw spectra of samples; S1: The purified spectra; LBP: Local Binary Patterns; HOG: Histogram of Oriented Gradient; GLCM: Gray-level co-occurrence matrix; BiLSTM: 1D-CNN-BiLSTM; SVM: Support vector machine.

Overall, the results indicated that the 1D-CNN-BiLSTM could classify healthy and damaged seeds. And the combination data of S1 + LBP + GLCM achieved the best performance. Moreover, the findings from this study highlighted the potential benefits of employing 1D-CNN-BiLSTM model in multi-source data processing and demonstrated their capability to classify samples effectively.

3.4. The classification results based on band ratio image

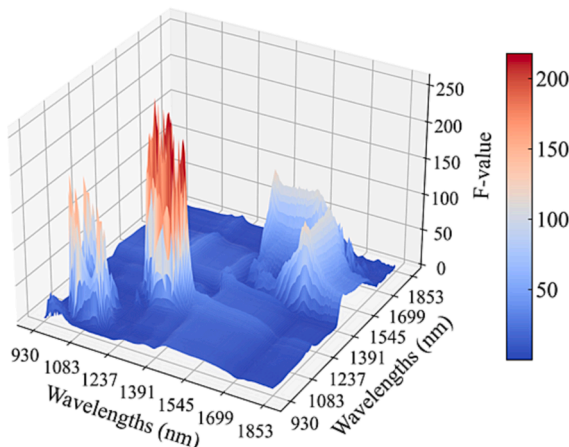
3.4.1. The ANOVA results for band ratio

For practical applications, utilizing fewer wavelengths in hyperspectral data means that a simpler and faster classification model could be developed. Additionally, it also can lead to cost savings during device development. Thus, the ANOVA was performed on the band ratio of the spectra data to select the optimal band ratio in this study. The advantage of this analysis method lies in its ability to identify significant differences in the band ratio among different classes within the data. The optimal band ratio exhibited the highest F-value. The results of ANOVA can be found in Fig. 3(a). It is clear that the optimal band ratio is 1160 nm / 1310 nm, which has the highest F-value. The threshold-based classification method could classify samples with different labels

based on the optimal band ratio. Fig. 3(b) shows the histogram of band ratios for different labels in the calibration set. By analyzing the calibration set data, the optimal classification threshold was calculated as 0.988. However, the classification accuracy was only 0.86 when applying the threshold value in prediction set. The result indicated that simply applying threshold segmentation on the optimal band ratio was not the best choice, and further processing was required.

The advantage of hyperspectral data lies in the combination of image and spectra information. Therefore, the optimal band ratio image could be obtained based on the optimal band ratio selected by ANOVA. It is clear that this approach compressed the band number to 2 for modeling, significantly improving model efficiency and reducing development costs. Moreover, the selected bands were highly correlated with labels and possessed strong mathematical interpretability. Then, the GLCM, LBP, and HOG textures were extracted from the optimal band ratio image. The SVM and 1D-CNN-BiLSTM models were established based on different combinations of these textures. The algorithm flow is illustrated in Fig. 2.

(a) 3D Heatmap of Band Ratios F-value



(b)

Histogram of Band Ratios for Different Labels in the Calibration Set

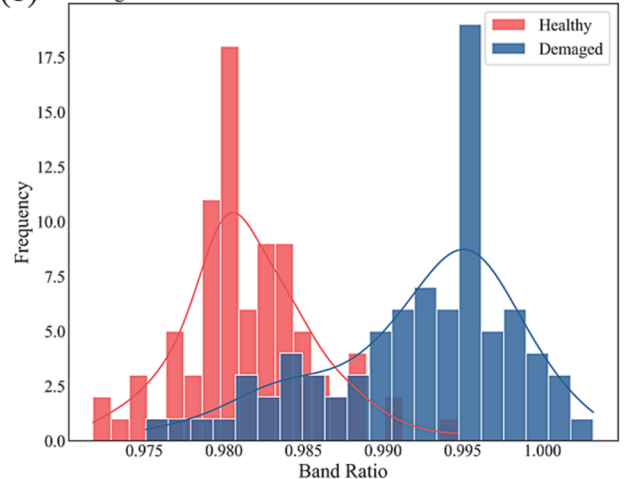


Fig. 3. (a) The result of ANOVA for band ratio and (b) the histogram of band ratios for different labels in the calibration set.

3.4.2. The classification results using texture features extracted from band ratio image

The Table 2 presents the classification results obtained from two different models, SVM and 1D-CNN-BiLSTM, using various texture features. Overall, whether using individual texture features or combining multiple texture features, the 1D-CNN-BiLSTM model consistently demonstrated superior performance compared to the SVM model. This highlights the advantage of the 1D-CNN-BiLSTM model in learning and combining relevant features from different sources, leading to improved discrimination ability.

Furthermore, it can be observed from the Table 2 that there is no significant difference in the results of the 1D-CNN-BiLSTM models established with different texture combinations. However, the complexity of the model and the number of features are also important criteria for evaluating the model's performance. It is essential to consider how to build a low-complexity model with fewer features in the modeling process. Therefore, considering all factors, the GLCM-1D-CNN-BiLSTM model achieved the best classification performance, with precision, recall, F1-score, and accuracy of 0.92, 1.00, 0.96, and 0.96, respectively. Only six texture features extracted from GLCM were used to establish the model. Furthermore, this model achieved the highest F1-score and accuracy among all models, indicating a better balance in the classification task by simultaneously considering precision and recall.

The above experimental results indicated that the 1D-CNN-BiLSTM model could classify healthy and insect-infested maize seeds using one-dimensional data. Its performance in multi-source data fusion was superior to the traditional machine learning method. Moreover, the classification performance of 1D-CNN-BiLSTM model was further improved after purifying the hyperspectral data using ANOVA method, achieving a classification accuracy of 0.96 with only two band images. Although this approach could achieve the classification goal, it still required the extraction of texture data from the images during the algorithm construction process. Therefore, it is preferable to directly use band ratio images to build a 2D-CNN model to simplify the model construction process.

3.4.3. The classification results based on 2D-CNN using band ratio image

The Table 3 shows the classification results based on 2D-CNN model using a band ratio image. It is evident that the 2D-CNN model could also achieve the classification of healthy and insect-infested maize seeds. However, in comparison to the 1D-CNN-BiLSTM model, this model's performance was slightly lower, with classification precision, recall, F1-score, and accuracy of only 0.97, 0.83, 0.89, and 0.90, respectively. This could be attributed to the 2D-CNN model's requirement for a larger dataset for construction, while the dataset used in this experiment was relatively small. Conversely, the 1D-CNN-BiLSTM model was more

suitable for small datasets.

However, the advantages of 2D-CNN models were quite apparent when compared to the traditional SVM model. It utilized only two band images and eliminated the texture data extraction process, achieving classification performance that is on par with the SVM model while simplifying the model construction process.

4. Conclusions

In this study, the hyperspectral data (930–1866 nm) and intelligent algorithms were used to explore the classification of healthy and insect-infested maize seeds. The 1D-CNN-BiLSTM and SVM models were applied to different combinations of spectra and texture features. The results demonstrated that the 1D-CNN-BiLSTM model outperformed SVM when dealing with multi-source data. The combination of spectra (S1) with LBP and GLCM achieved the best classification performance, highlighting the advantage of 1D-CNN-BiLSTM in handling complexities and interactions among texture features. The F1-score and accuracy of the S1 + LBP + GLCM + 1D-CNN-BiLSTM model were both 0.96. Moreover, the optimal band ratio of 1160 nm/ 1310 nm was identified by using the ANOVA method. Moreover, the 2D-CNN model also achieved classification with the F1-score and accuracy of 0.89 and 0.90; its performance was slightly lower than the 1D-CNN-BiLSTM model. However, the 2D-CNN also showcased advantages over other models, using fewer bands and eliminating the texture data extraction step. The 1D-CNN-BiLSTM model, using texture features from the band ratio image, demonstrated superior classification performance with only two band images. The GLCM + 1D-CNN-BiLSTM model obtained the best results with an F1-score and accuracy of 0.96, respectively. This approach efficiently reduced the number of band and model complexity, making it a promising choice for practical applications. Therefore, the GLCM + 1D-CNN-BiLSTM model was regarded as the best performance model.

CRediT authorship contribution statement

Zheli Wang: Writing – original draft, Software, Methodology. **Shuxiang Fan:** Software, Methodology, Conceptualization. **Ting An:** Software, Methodology, Investigation. **Chi Zhang:** Software, Methodology. **Liping Chen:** Writing – review & editing, Supervision, Conceptualization. **Wenqian Huang:** Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

Table 2

The classification results of SVM and 1D-CNN-BiLSTM model based on texture features from band ratio image.

Data	Model	Calibration set				Prediction set			
		Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy
GLCM	SVM	0.90	0.85	0.87	0.88	0.88	0.86	0.87	0.88
	BiLSTM	0.99	0.95	0.97	0.97	0.92	1.00	0.96	0.96
LBP	SVM	0.71	0.65	0.68	0.69	0.78	0.71	0.75	0.76
	BiLSTM	1.00	0.99	0.99	0.99	0.94	0.86	0.90	0.90
HOG	SVM	0.83	0.79	0.81	0.81	0.73	0.69	0.71	0.72
	BiLSTM	0.93	1.00	0.98	0.98	0.92	0.94	0.93	0.93
HOG + GLCM	SVM	0.84	0.86	0.85	0.85	0.76	0.74	0.75	0.76
	BiLSTM	0.92	1.00	0.96	0.96	0.89	0.97	0.93	0.93
HOG + LBP	SVM	0.86	0.85	0.85	0.85	0.79	0.74	0.76	0.78
	BiLSTM	0.92	1.00	0.96	0.96	0.85	1.00	0.92	0.92
GLCM + LBP	SVM	0.93	0.91	0.92	0.92	0.86	0.89	0.87	0.88
	BiLSTM	0.93	1.00	0.97	0.96	0.91	0.91	0.91	0.92
GLCM + LBP + HOG	SVM	0.97	0.99	0.98	0.98	0.86	0.86	0.86	0.86
	BiLSTM	0.99	0.99	0.99	0.99	0.92	0.97	0.94	0.94

Abbreviation: LBP: Local Binary Patterns; HOG: Histogram of Oriented Gradient; GLCM: Gray-level co-occurrence matrix; BiLSTM: 1D-CNN-BiLSTM; SVM: Support vector machine.

Table 3

The classification results based on 2D-CNN using band ratio image.

Data	Model	Calibration set				Prediction set			
		Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy
Band ratio image	2D-CNN	0.95	0.88	0.91	0.92	0.97	0.83	0.89	0.90

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the financial support provided by the National Natural Science Foundation of China (31801262), Young Elite Scientists Sponsorship Program by CAST(2019QNRC001).

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